CO544: Machine Learning and Data Mining
Lab 05: Text Classification and Performance Analysis

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TASK 1: Find data preprocessing steps other than mentioned above.

Apart from the steps mentioned in the given code, the following preprocessing techniques can be used to further clean and improve the text data for machine learning models:

➔ **1. Removing Stop Words:-**Common words like "the", "is", "at", and "which" appear frequently in language but do not carry significant meaning for sentiment analysis. Removing these words helps the model focus only on the important words that determine whether a review is positive or negative.

➔ **2. Removing HTML Tags:-**Since this dataset is collected from movie review websites (IMDb), the raw text may contain HTML formatting tags like
 or text links. These tags act as noise and should be removed using regular expressions before training the model.

➔ **3. Stemming:-** As an alternative to Lemmatization, Stemming can be used to chop off the endings of words to bring them to their base form (ex:- "studying", "studied", and "studies" all become "studi"). It is a faster but less accurate rule-based method compared to Lemmatization.

➔ **4. Handling Emoticons and Emojis:-** In movie reviews, users often express emotions using punctuation marks like :) or :(. Converting these emoticons into meaningful words (ex:- replacing :) with "happy") ensures that the sentiment information they carry is not lost during cleaning.

TASK 2: Discuss advantages and disadvantages of the Bag of Words model.

Advantages:-

- ➔ **Very Simple:-** It is very easy to understand and simple to implement.
- ➔ **Works Well for Classification:-** It works efficiently for text classification tasks like spam detection and basic sentiment analysis.
- ➔ **Fixed Output Size:-** No matter how long the text is, it converts every document into a fixed size numerical format (array).

Disadvantages:-

- ➔ **Loses Word Order:-** It ignores the order or sequence of words. For example, "not good" and "good, not" will look exactly the same to the model.
- ➔ **Sparsity (More Zeros):-** The resulting matrix contains a lot of zeros because most words in the vocabulary do not appear in every review. This wastes computer memory.
- ➔ **Loses Context (Semantics):-** It treats words as completely independent. It cannot understand synonyms (e.g., it does not know that "huge" and "large" mean similar things).

TASK 3: Train a Random Forest model, a Support Vector Machine model and a Naive Bayesian classifier. Compare the accuracies and other performance measures (precision, recall, F1-score, confusion matrix) of all four models including the Logistic Regression model. What is the best model? Justify your answer based on these measures.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.8200	0.8213	0.8213	0.8200
Random Forest	**0.8275**	**0.8279**	**0.8283**	**0.8275**
Support Vector Machine (SVM)	0.8100	0.8102	0.8107	0.8100
Naive Bayes	0.7450	0.7461	0.7430	0.7434

Conclusion:- Selection of the Best Model

Based on the experimental results, the **Random Forest** model is selected as the best performing model for this movie review text classification task.why?

- ➔ **Highest Overall Accuracy:-** The **Random Forest** model achieved the highest accuracy of **82.75%**, outperforming Logistic Regression (82.00%), SVM (81.00%), and Naive Bayes (74.50%).
- ➔ **Best Balance (Highest F1-Score):-** Since accuracy alone is not always enough, we look at the F1-Score. The Random Forest model achieved the highest F1-Score of **0.8275**. This indicates that the model has an excellent balance between **Precision** (avoiding false positives) and **Recall** (avoiding false negatives).
- ➔ **Robustness against Naive Bayes:-** The Naive Bayes classifier showed the lowest performance (74.50% accuracy). This is likely because Naive Bayes assumes all words are completely independent, whereas Random Forest captures non linear relationships and interactions between words much better.
